Empirical Methods in Finance

Do oil price increases have a stronger negative effect on equity returns during periods of economic weakness, and does this effect differ between developed and emerging markets ?

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Introduction

Oil is one of the most closely followed commodities in global financial markets. Since Hamilton (1983) showed that oil price shocks were behind most post-war recessions in the United States, researchers have tried to understand how oil price movements affect equity markets. The evidence remains mixed, partly because not all oil price increases carry the same meaning: some reflect supply disruptions while others are driven by strong global demand, with potentially opposite effects on financial markets.

This paper addresses the following question: **Do oil price increases have a stronger negative effect on equity returns during periods of economic weakness, and does this effect differ between developed and emerging markets?** Both dimensions matter in practice. When the economy is already struggling, firms and households are more sensitive to rising energy costs, which could amplify the impact of oil shocks on stock prices. Emerging markets may also react differently from developed ones, given their greater dependence on energy imports and less diversified economies.

To answer this question, we use two complementary approaches: a static OLS model extended to account for asymmetric and state-dependent effects, and a VAR model that tracks how oil shocks propagate over time. Our results suggest that oil price increases mostly reflect demand conditions rather than supply disruptions, and do not appear to have a stronger negative effect during downturns. We also find little difference between developed and emerging markets.

Literature Review

The relationship between oil prices and economic activity has been extensively investigated. Early contributions by Hamilton (1983) show that positive oil price shocks have been a major driver of economic recessions in the United States. Then many studies have analyzed the relationship between oil prices and stock market returns. A contribution of Sadorsky (1999) shows that oil price shocks affect both stock returns and economic activity, with positive shocks generally having a stronger impact than negative ones. However, the empirical evidence remains mixed. While some studies confirm a significant relationship (Basher et al., 2012; Jones and Kaul, 1996; Kilian and Park, 2009), others report weaker or insignificant effects (Aspergis and Miller, 2009; Miller and Ratti, 2009).This suggests that the impact of oil prices may depend on economic conditions and model specifications. Kilian (2009) further argues that not all oil price shocks are alike, differentiating between supply and demand driven shocks with different macroeconomic consequences.

Some literature focuses on asymmetric effects. Several studies argue that increases in oil prices have a stronger impact than decreases, reflecting the role of oil as a negative supply shock. Using advanced econometric techniques such as EGARCH models, Narayan and Narayan (2012) show that oil price shocks have persistent and asymmetric effects on volatility. Similarly, Chiou and Lee (2009) find that unexpected oil price fluctuations have nonlinear and asymmetric effects on stock returns.These findings indicate that financial markets do not react symmetrically to oil price movements and that the direction of the shock plays an important role.

The dynamic interaction between oil prices and financial markets has been studied using multivariate time-series frameworks. Vector Autoregressive (VAR) models are commonly used to capture the interdependence between variables and the propagation of shocks over time.For instance, Huang et al. (1996) use a VAR approach to analyse the relationship between oil futures and stock returns, finding limited effects on broad market indices. More recent studies extend this analysis using more advanced econometric approaches, including VARMA-GARCH, wavelet, and copula models (Filis et al., 2011; Fratzscher et al., 2013; Chang et al., 2013), highlighting the time-varying and nonlinear nature of the oil–stock market relationship.

The literature also suggests that the impact of oil price shocks varies across countries and economic structures. Emerging markets are often found to be more sensitive to oil price fluctuations due to their higher dependence on energy consumption and less diversified economic systems.However, most studies analyze developed and emerging markets separately, rather than comparing them within a unified and consitent empirical framework.

Furthermore, the impact of oil price shocks may depend on the broader economic environment. Since oil price increases have been identified as important drivers of economic downturns (Hamilton, 1983), their effects on financial markets are likely to be amplified during periods of economic weakness. In such periods, firms and households are more vulnerable on equity returns. Despite this important insight, relatively few studies explicitly combine the analysis of oil price asymmetry, economic conditions, and cross-market differences.

Data

Financial Dataset (Daily Frequency)

Our financial data cover the period from January 1990 to March 2026. All price series are transformed into log-returns, defined as:

$$r_t = 100 \times (\log P_t - \log P_{t-1}) \tag{1}$$

For the US 10-year Treasury yield, which is expressed in levels, we compute first differences instead of log-returns, as yields do not represent price processes.

Table 1: Financial variables

Variable	Role	Justification
Brent crude oil	Main oil variable	Global benchmark for oil price
MSCI World	Developed markets	Proxy for global equity performance
MSCI EM	Emerging markets	Higher sensitivity to oil price fluctuations
US 10-year rate	Interest-rate control	Controls for monetary environment

Macroeconomic Dataset (Monthly Frequency)

The macroeconomic data cover the same sample period as the financial data. To ensure that the oil data macroeconomic variables are based on the same frequency, we calculate monthly oil returns by aggregating the daily log-returns within each month.

Table 2: Macroeconomic variables

Variable	Role	Justification
Manufacturing ISM	Economic conditions	Captures expansion/contraction (threshold: 50)
CFNAI	Economic activity	Recession proxy (threshold: -0.7)
Industrial Production	Economic activity	Alternative measure of real output

Methodology

This study employs a two-step empirical strategy that combines static and dynamic approaches. Firstly, we estimate a baseline OLS model to capture the linear relationship between changes in oil prices and stock market returns. The OLS regression is then extended to explore two additional dimensions. We examine asymmetric effects by decomposing changes in oil prices into positive and negative components and then we examine state-dependent effects by introducing a recession dummy variable based on the CFNAI index, which allows us to assess whether the impact of oil shocks is amplified during periods of economic weakness. Robustness tests are conducted using alternative macroeconomic variables to ensure the stability of the results of the macroeconomic extension. Secondly, we use a vector autoregressive (VAR) model following Huang et al. (1996) to analyze the dynamic transmission of oil shocks over time. This approach allows us to capture the feedback effects between oil prices, stock markets, and interest rates, and to study how shocks propagate through the system.

Preliminary Diagnostics

Descriptive statistics: We first compute for all return series: mean, variance, skewness, and kurtosis. Skewness measures the asymmetry of the distribution, while excess kurtosis captures the presence of fat tails relative to the normal distribution. We then assess the normality of return series using the Jarque-Bera test, defined as:

$$JB = T\left(\frac{\hat{S}^2}{6} + \frac{(\hat{K}-3)^2}{24}\right) \tag{2}$$

where $\hat{S}$ and $\hat{K}$ denote sample skewness and kurtosis, respectively.

The results show that all series are negatively skewed and have positive excess kurtosis (see Appendix Figure 2). Oil returns show the strongest deviation from normality (skewness of -0.95 and excess kurtosis of 18.7), which reflect the presence of rare but extreme price movements. The Jarque–Bera confirms that the null hypothesis of normality can be rejected at the 5% level for all series.

Next, we examine the stationarity properties of the series using two complementary tests. The Augmented Dickey-Fuller (ADF) test has the null hypothesis of a unit root, while the KPSS test assumes stationarity under the null. Combining both tests provides a more robust assessment: a series is considered stationary when the ADF rejects the unit root hypothesis and the KPSS test fails to reject stationarity.

All return series appear stationary according to both tests, except for the Industrial production variable (see Appendix Table 2). However, since this variable is only used as a binary dummy in the robustness checks rather than directly in the regressions, this does not affect our estimation.

We further analyze time dependence by examining serial correlation in both returns and squared returns using autocorrelation functions (ACF) and Ljung-Box tests. The Ljung-Box statistic is defined as:

$$Q_p = T(T+2)\sum_{j=1}^{p}\frac{\hat{\rho}_j^2}{T-j} \tag{3}$$

The Ljung-Box test rejects the null hypothesis of no autocorrelation for all series. Significant autocorrelation in returns indicates predictability in the mean, while autocorrelation in squared returns is interpreted as evidence of volatility clustering, consistent with the stylized facts documented by Cont (2001).This does not bias OLS estimates but may affect the standard errors, so we use heteroskedasticity-robust standard errors in the regressions.

We also compute the correlation matrix across all return series to assess the degree of co-movement between oil prices and financial markets. Oil returns show moderate positive correlations with MSCI World 0.20 and MSCI EM 0.17 (see Appendix Figure 3). This suggests that the relationship between oil and equity markets may not be fully captured by a simple linear specification, which motivates the use of conditional and asymmetric models.

Oil return decomposition: We decompose oil returns into positive and negative components:

$$r^{+}_{oil,t} = \max(r_{oil,t},\, 0), \qquad r^{-}_{oil,t} = \min(r_{oil,t},\, 0) \tag{4}$$

This decomposition allows us to investigate potential asymmetries in the effects of oil price shocks by comparing the behavior of positive and negative changes.

Baseline OLS Model

All models are estimated separately for MSCI World and MSCI Emerging Markets in order to assess whether the relationship between oil and equity differs across different market types.

As a first step, we estimate two baseline OLS regressions at the daily frequency to measure the simultaneous effect of oil price changes on equity returns:

$$r_{world,t} = \alpha + \beta_1 r_{oil,t} + \beta_2 r_{rate,t} + \varepsilon_t \tag{5}$$

$$r_{em,t} = \alpha + \beta_1 r_{oil,t} + \beta_2 r_{rate,t} + \varepsilon_t \tag{6}$$

where $r_{world,t}$ and $r_{em,t}$ denote daily log-returns of MSCI World and MSCI Emerging Markets, respectively, $r_{oil,t}$ denotes Brent oil returns, and $r_{rate,t}$ represents changes in the US 10-year Treasury yield.

To account for potential heteroskedasticity and autocorrelation in the residuals, as mentioned previously, we use Newey-West HAC standard errors.

In addition, to test whether oil price increases and decreases have different effects on equity returns, we use oil return decomposition components:

$$r^{+}_{oil,t} = \max(r_{oil,t},\, 0), \qquad r^{-}_{oil,t} = \min(r_{oil,t},\, 0) \tag{7}$$

The model is then extended as follows:

$$r_{world,t} = \alpha + \beta_1 r^{+}_{oil,t} + \beta_2 r^{-}_{oil,t} + \beta_3 r_{rate,t} + \varepsilon_t \tag{8}$$

To test whether the effect of oil shocks is amplified during periods of economic weakness, we introduce a recession dummy based on the CFNAI index:

$$\text{Recession}_t = \mathbf{1}(\text{CFNAI}_t < -0.7) \tag{9}$$

The threshold of -0.7 follows(Brave, 2009), who shows that a CFNAI-MA3 below this level has historically predicted recession months with 86 % accuracy since 1967. We then estimate the following interaction model at the monthly frequency:

$$r_{world,t} = \alpha + \beta_1 r_{oil,t} + \beta_2 \text{Recession}_t + \beta_3 (r_{oil,t} \times \text{Recession}_t) + \varepsilon_t \tag{10}$$

where β_3 captures the additional effect of oil price variation during recession periods.

We also run three robustness checks to test whether our main result still holds under alternative measures of economic weakness. First, we replace the CFNAI dummy with an Industrial Production (IP) dummy which is equal to 1 when IP is below its sample median. We then use an ISM Manufacturing dummy, equal to 1 when ISM is below 50, which is the standard threshold separating expansion from contraction. For the last check, we use the continuous CFNAI value in the interaction rather than a binary dummy.

VAR Model

To analyze the dynamic transmission of oil price shocks on the equity market, we estimate two separate VAR models. The first model focuses on developed markets, while the second focuses on emerging markets. The systems are specified as follows:

$$\mathbf{y}_{1,t} = \mathbf{c} + A_1 \mathbf{y}_{1,t-1} + \cdots + A_p \mathbf{y}_{1,t-p} + \mathbf{u}_t \tag{11}$$

$$\mathbf{y}_{2,t} = \mathbf{c} + A_1 \mathbf{y}_{2,t-1} + \cdots + A_p \mathbf{y}_{2,t-p} + \mathbf{u}_t \tag{12}$$

where

$$\mathbf{y}_{1,t} = \begin{pmatrix} r_{oil,t} \\ r_{world,t} \\ r_{rate,t} \end{pmatrix}, \qquad \mathbf{y}_{2,t} = \begin{pmatrix} r_{oil,t} \\ r_{em,t} \\ r_{rate,t} \end{pmatrix}$$

Oil is ordered first in the Cholesky decomposition, reflecting the assumption that oil price shocks can affect financial markets at the same time, while financial markets do not affect oil prices within the same period.

The optimal lag length p is selected using standard information criteria. While the BIC and HQ criteria suggest a short lag specification, the AIC and FPE indicate a longer lag structure. Both specifications are estimated, but greater emphasis is placed on the higher lag model, as it captures more of the underlying dynamics and provides more informative impulse response functions. We then perform Granger causality tests to assess whether past oil returns contain significant predictive information for current equity returns.

Finally, we compute orthogonalized impulse response functions (IRFs) based on the Cholesky decomposition in order to trace the response of equity markets to a one-standard-deviation oil price shock over time. Confidence bands are computed using the Monte Carlo simulation with 300 replications at the 95 % level.

Macroeconomic Extension

To further investigate the transmission of oil price shocks to the real economy, we estimate a third VAR model at a monthly frequency:

$$\mathbf{y}_{3,t} = \begin{pmatrix} r_{oil,t}^{monthly} \\ \text{Manufacturing ISM}_t \\ \text{CFNAI}_t \end{pmatrix}$$

The same empirical framework is applied, including lag selection using information criteria, Granger causality tests, stability diagnostics, and impulse response function analysis.

This lower monthly frequency is not only dictated by data availability, but also allows us to appropriately track the slower, more persistent transmission of energy prices to the real economy.

Forecasting

To assess the predictive power of oil prices, we split the sample into a training period (1990-2015) and an out-of-sample test period (2016-end of sample). The VAR models are estimated on the training sample only, and forecasts are generated for the test period.

We apply this evaluation to two models: the monthly VAR forecasting MSCI World returns, and the monthly VAR (y3) forecasting macroeconomic conditions. The predicted values are then plotted against realized values to visually assess whether oil price dynamics provide genuine predictive power.

Forecasting performance is evaluated using the Root Mean Squared Error (RMSE) :

$$\text{RMSE} = \sqrt{\frac{1}{H}\sum_{h=1}^{H}(y_{t+h} - \hat{y}_{t+h})^2}$$

We then compare each RMSE model to the standard deviation of the series to assess whether the model predicts better than the historical average.

Results

Baseline Models

Our baseline model shows a positive and significant relationship between oil and equity returns at the daily frequency (see Appendix Table 1). A 1% increase in oil prices is associated with a 0.077% increase in MSCI world returns and a 0.079% increase for MSCI Emerging Markets returns, both significant at the 1% level. The coefficients are similar across the two markets, suggesting that the immediate transmission of oil price shocks does not really differ between developed and emerging markets at the daily frequency. This finding does not provide evidence of stronger impact in emerging markets. Interest rate changes are also positive and significant across both specifications, consistent with the idea that rising rates tend to coincide with periods of economic expansion. The R-squared is low (6.7% and 3.8%), but this is expected for daily return regressions.

Asymmetric Effects

The model reveals that positive and negative changes in oil prices have different coefficient magnitudes. A 1% increase in the oil price is associated with a 0.056% increase in MSCI World Index returns and a 0.059% increase in MSCI Emerging Markets Index returns. However, a 1% decline in oil prices is associated with a 0.095% decline in MSCI World returns and a 0.097% decline in MSCI Emerging Markets returns. The coefficient suggests that negative changes in oil prices have a stronger impact on stock market returns than positive changes. The effect is similar in both developed and emerging markets. The stronger impact of negative oil price movements may reflect episodes of global economic weakness, during which both oil prices and stock markets deteriorate. A formal test fails to reject the null of equal coefficients ($t = -1.17$ for World, $t = -1.01$ for EM), which means we cannot formally conclude that the effects are statistically asymmetric at the 5% level. The interest rate control variable is positive and significant, confirming its importance in the model. But, given the static nature of the OLS specification, we cannot establish the dynamic transmission of oil shocks, which justifies the use of the VAR framework in the following section.

Macro Interaction Model

At the monthly frequency, the oil coefficient is 0.091 for MSCI World and 0.139 for MSCI Emerging Markets. During recession periods, the total effect of oil increases to 0.151 and 0.223 respectively. This suggests that oil shocks have a stronger impact on equity markets in bad times. However, the results have to be interpreted with caution because the interaction term is not statistically significant ($t = 0.59$ for MSCI World and $t = 0.60$ for MSCI emerging markets).

The recession indicator variable is negative in itself, but is not statistically significant for both markets, suggesting that periods of recession do not have a systematic independent effect on stock returns beyond the oil channel. For that, the interaction model provides no evidence that the impact of oil price changes on stock market returns differs across economic regimes. This suggests that the effect of oil price changes on stock markets does not appear to be amplified during periods of economic slowdown. In other words, based on our model, there is no statistical evidence supporting such effects. The lack of significance of the interaction term can be explained by several factors. First, periods of recession account for only a small fraction of the sample (2001, 2008–2009, and 2020, or approximately 8%), which limits the statistical power of the test. Finally, the fact that the CFNAI is a U.S. index, while the proxy for oil is a global index and market returns are for developed and emerging markets may introduce some biases like geographical mismatch.

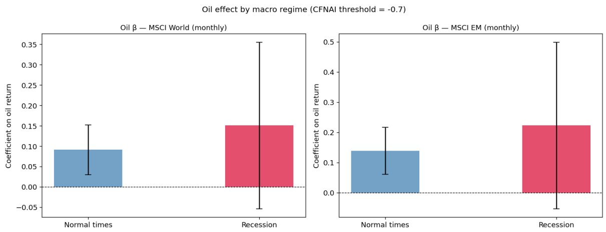

Figure 1: Oil effect by regime

OLS Robustness Checks

The oil coefficient stays positive and significant in all specifications, so the oil-equity relationship holds regardless of the macro indicator that is used. However, the interaction term gives different results depending on how economic weakness is measured. With the IP dummy ($t = -1.31$ for World and $t = -1.16$ for EM) and the ISM dummy ($t = -0.75$ for World and $t = -0.49$ for EM), the interaction is not significant and actually goes in the opposite direction to what we expected. The strongest result is the continuous CFNAI specification, where the interaction is significant for both markets ($t = -2.12$ for World and $t = -2.51$ for EM). This supports our hypothesis that oil price shocks have a stronger effect on equity markets when the economy is weak, with clearer evidence than when using the binary CFNAI dummy.

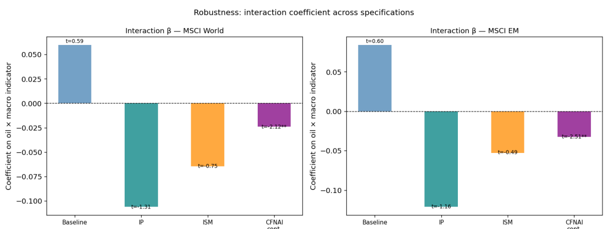

Figure 2: Interaction coefficient across specification

VAR Model

We estimate two VAR models to analyze the dynamic transmission of oil price shocks to stock markets. One for developed markets and the other for emerging markets. The lag selection criteria yield divergent but consistent results. While the BIC and HQ suggest a lag of $p = 1$, the AIC and FPE indicate a larger dynamic structure $p = 7$. We therefore estimate both cases to assess the sensitivity of the results to lags.

Granger causality tests show that oil returns contain significant predictive information for stock market returns in both markets. At $p = 1$, the relationship flows from oil to stock markets, which is consistent with our Cholesky ordering assumption, where oil is placed first as the most exogenous variable. Oil is a Granger-cause for the MSCI World ($F = 4.904$, $p = 0.027$) and the MSCI EM ($F = 5.953$, $p = 0.015$), while the reverse is not supported at the 5% level. The higher F-statistic for emerging markets is consistent with their greater dependence on oil.

At $p = 7$, stock markets are also Granger-causes for oil (World: $p = 0.002$, EM: $p = 0.036$), suggesting that stock markets may also contain predictive information for oil prices over longer horizons. This highlights the horizon-dependent nature of the oil-stock relationship and calls for caution when interpreting oil as strictly exogenous.

Diagnostic tests confirm that all VAR systems are stable. Residual autocorrelation is present with $p = 1$ (Ljung-Box p-values ranging from 0.000 to 0.023), indicating that the model does not fully capture the underlying dynamics. This is consistent with the volatility clustering identified earlier and supports our choice of Newey-West standard errors in the OLS models.

In contrast, the lag $p = 7$ significantly improves the diagnostics: autocorrelation is largely resolved for the oil and interest rate residuals, with marginal persistence in MSCI World returns ($p = 0.038$). This leads us to suggest that the lagged structure provides a more adequate representation of the data generation process. Based on this, greater weight is given to the results obtained under the $p = 7$ lag.

Impulse Response Functions (Daily VAR)

Impulse response functions allow for a quantitative assessment of the dynamic effects of oil price shocks on stock markets. Under a lag of $p = 7$, an oil price shock equivalent to one standard deviation leads to an immediate increase of approximately 0.18% in the returns of both the MSCI World Index and the MSCI Emerging Markets Index. This effect is statistically significant at the time of the impact, with confidence intervals remaining above zero. But the response declines rapidly and becomes statistically insignificant after the first period, with confidence intervals crossing zero. Over the following 5 to 7 days, both markets exhibit fluctuations around zero, ranging from -0.03% to +0.02%. This indicates short-term adjustments in the stock markets following the initial shock, but does not provide evidence of persistent or economically significant dynamics. The effect of oil price shocks on stock markets is therefore short-lived and absorbed within a few periods. This rapid mean-reversion is consistent with the efficient market hypothesis, whereby oil prices news is quickly incorporated into asset prices on a daily basis.

These responses are very similar between developed and emerging markets, with only marginal differences in magnitude, which is surprising given the higher value of the Granger causality F-statistic for emerging markets. This suggests that while oil returns are more predictive for emerging markets, the dynamic response to a structural shock is not significantly different. The response of interest rates to oil price shocks is weak and statistically insignificant across all horizons. This indicates that monetary conditions do not adjust significantly to oil price movements at the frequency under consideration.

The impulse response analysis confirms that oil price increases have a positive but short-lived effect on stock markets, with no substantial difference between developed and emerging markets. The positive direction of the response is consistent with a demand-driven interpretation of oil price fluctuations, as suggested by Kilian (2009).

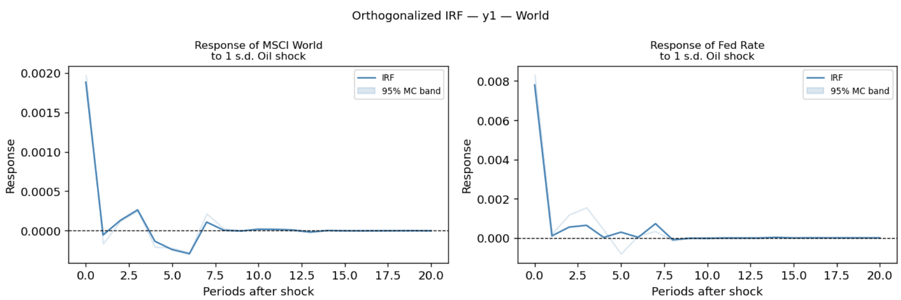

Figure 3: Irf World with p=7

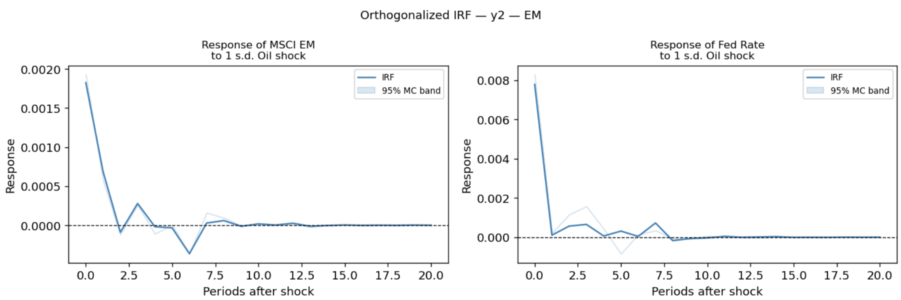

Figure 4: Irf Emerging markets with p=7

Impulse Response Functions (Monthly Macro VAR)

The macroeconomic extension estimates a monthly VAR including oil returns, the U.S. Manufacturing ISM index, and the U.S. CFNAI. For the Lag selection criteria BIC and

HQ select $p = 1$, while AIC and FPE select $p = 2$. Both lags are estimated, with greater emphasis on the higher lag model since it captures all dynamics in the system, although overfitting can remains given the nature of the criteria.

Granger causality test show a strong and robust transmission from oil prices to macroeconomic variables. Oil returns significantly Granger-cause both Manufacturing ISM and CFNAI across different lags, with particularly large F-statistics for CFNAI ($F = 70.21$ at $p = 1$ and $F = 31.11$ at $p = 2$). Manufacturing ISM Granger-causes CFNAI, suggesting that changes in industrial activity propagate to broader economic conditions.

Impulse response functions provide further insights. Following a one-standard-deviation oil shock, Manufacturing ISM rise rapidly, peaking at around 0.8 within 2 to 3 months, and remains positive for more than 15 to 20 months. These show that positive oil price shocks are associated with short to medium term improvements in industrial activity which is consistent with a demand-driven interpretation of oil price movements.

In contrast, CFNAI increases on impact, rise at around 0.4 within the first month, but quickly becomes statistically insignificant. Under the higher lag, the response even turns temporarily negative before stabilizing around zero. This suggests that while oil shocks may coincide with short-term improvements in activity, they do not generate persistent effects on overall economic conditions.

These results suggest that oil price shocks are associated with short term fluctuations in economic activity. This finding provides important context for our research question: if oil price increases do not generate persistent economic downturns, their amplified impact on equity returns during recessions is less likely to be detected.

Taken together, the macroeconomic analysis shows that oil price shocks transmit to the U.S. real economy, but the effects are short lived and do not translate into sustained macroeconomic weakness, which helps explain the absence of strong state dependent effects in equity markets.

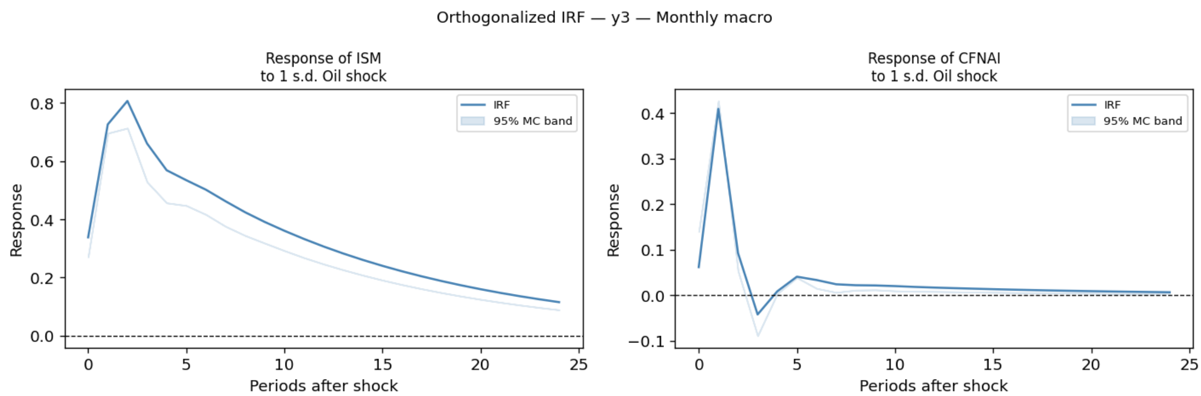

Figure 5: Irf Macro with p=2

Forecasting

The out-of-sample results show that the VAR models have limited forecasting ability. For MSCI World daily returns, the RMSE is 0.0095, almost identical to the historical standard deviation of 0.0094. This means that the VAR does no better than simply forecasting the historical mean. The same pattern holds for the monthly macro VAR, with RMSE/std ratios of 1.07 for ISM and 1.72 for CFNAI. The poor performance on CFNAI is largely driven by the COVID-19 shock of April 2020, which is impossible to forecast with a standard VAR. Overall, while oil price movements are statistically related to equity and macro variables in-sample, this does not translate into useful out-of-sample predictions. This result is consistent with the findings of Welch and Goyal (2008), who show that predictive models of financial returns generally fail to outperform the historical mean in out-of-sample evaluations.

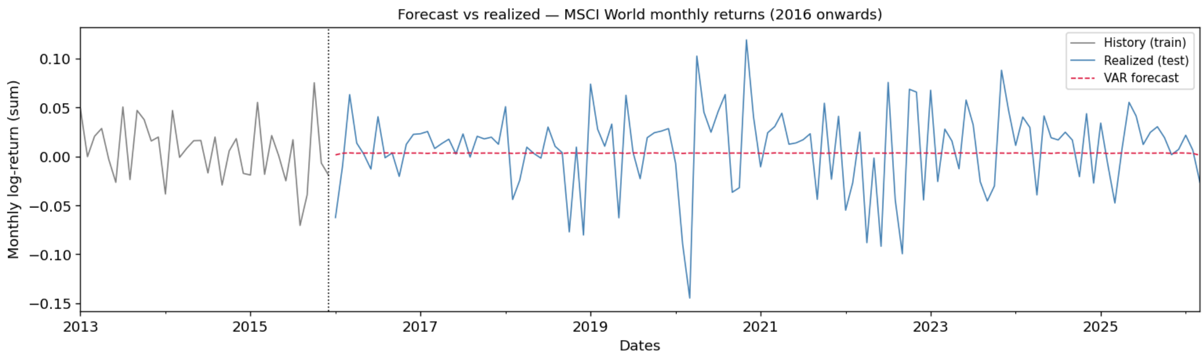

Figure 6: MSCI World Forecast

Limitations

Our study presents some limitations. First, our analysis relies on full-sample estimation, which may mask heterogeneity across different oil price episodes. Future research could focus on specific oil shock periods (major oil crises or geopolitical events) in order to assess whether the transmission of oil shocks differs across episodes.

Second, while the macroeconomic analysis is conducted at a monthly frequency, the financial analysis relies on daily data. High-frequency data may introduce substantial noise and obscure more persistent relationships. Future research could extend the analysis by applying the VAR framework to lower-frequency financial data (weekly or monthly returns) to better capture medium to long-term dynamics.

Another limitation of this study is the presence of global shocks that simultaneously affect oil prices and financial markets. Major events such as the 2008 financial crisis and the COVID-19 pandemic were not driven by oil market dynamics but had significant impacts on both oil prices and equity markets. As a result, the VAR framework may attribute part of the co-movement between oil prices and financial markets to oil shocks, while both variables may in fact be responding to common underlying macroeconomic shocks. This issue reflects the presence of omitted common factors.

Additionally, the -0.7 threshold is actually defined for the three-month moving average of the CFNAI, not the monthly index. We apply it directly to the monthly series for simplicity, which may slightly affect the dating of recessions. Furthermore, the small number of recession months in our sample (35 out of 434) limits the precision of the interaction estimates. As shown in the robustness section, the interaction results are also sensitive to the choice of macro indicator. While the continuous CFNAI supports our hypothesis, the IP and ISM dummy specification give insignificant and opposite-signed coefficients. This sensitivity reflects a deeper limitation: our conclusions depend on which indicator is used to capture economic weakness.

The OLS and VAR frameworks assume linear relationships between variables. However, the relationship between oil prices and financial markets may be nonlinear, particularly during extreme market conditions. For instance, a large oil shock during a crisis may have a disproportionately stronger effect than a small shock in normal times. More flexible approaches such as SVAR or GARCH-based models could better capture these dynamics.

Finally, the CFNAI is also a US-specific index, while the equity indices cover global developed and emerging markets. Periods of US economic weakness do not necessarily coincide with downturns in global or emerging market economies, which may partly explain why no strong state-dependent effects are found.

Conclusion

This paper examines whether oil price increases have a stronger negative impact on equity returns during periods of economic weakness, and whether this effect differs between developed and emerging markets. Using both OLS and VAR approaches, we find no robust evidence supporting this hypothesis.

At the daily frequency, oil price increases are positively associated with equity returns, suggesting that oil prices mainly reflect global demand conditions rather than negative supply shocks. Second, although negative oil price changes appear to have a stronger effect on equity returns, this asymmetry likely captures periods of weak economic activity during which both oil prices and stock markets decline.

Importantly, we find no robust evidence that the effect of oil price shocks is amplified during periods of economic weakness. While one specification using a continuous macroeconomic indicator provides some support, the overall results do not confirm strong state-dependent effects. In addition, the results are very similar for developed and emerging markets.

The VAR analysis shows that the impact of oil price shocks on equity markets is short-lived, with a small positive effect that disappears within a few days. At the macroeconomic level, oil prices shocks have a positive and persistent effect on manufacturing activity, but only a weak and temporary effect on economic activity.

Overall, our results suggest that the relationship between oil prices and equity returns is largely driven by common macroeconomic conditions rather than a direct causal mechanism. Oil price increases do not appear to have a stronger negative effect during periods of economic weakness, nor do they affect developed and emerging markets differently. Future research could extend this work by identifying structural supply and demand shocks separately, or by using time-varying VAR models to test whether the relationship between oil and equity markets has changed over time.

Appendix

Figure 1: Time Series of Financial Returns

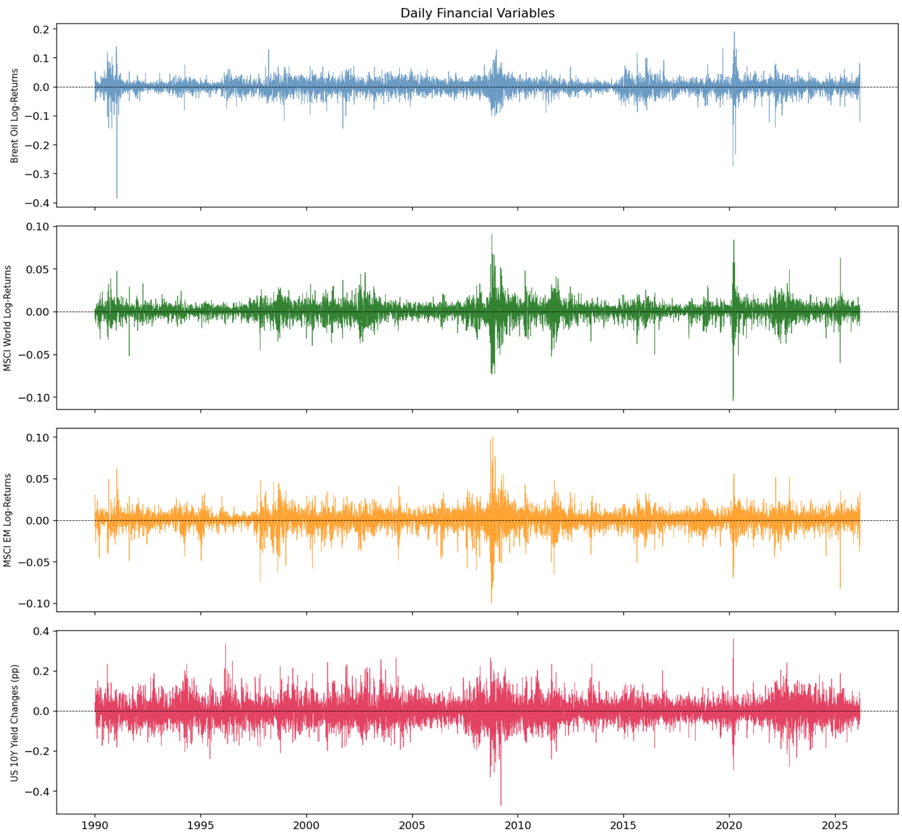

Figure 2: Skewness and Excess Kurtosis

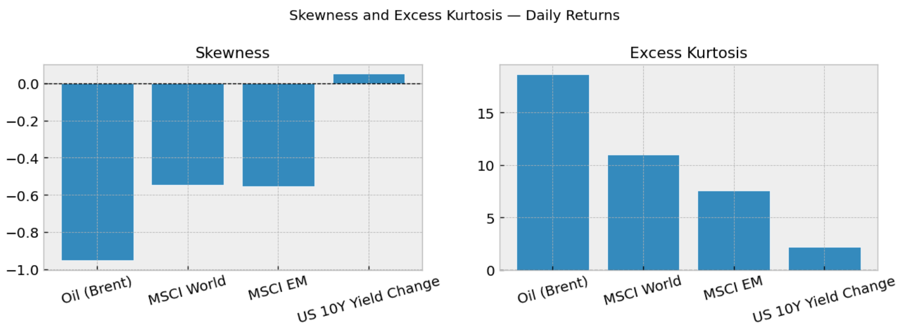

Figure 3: Cross-Correlation Matrix

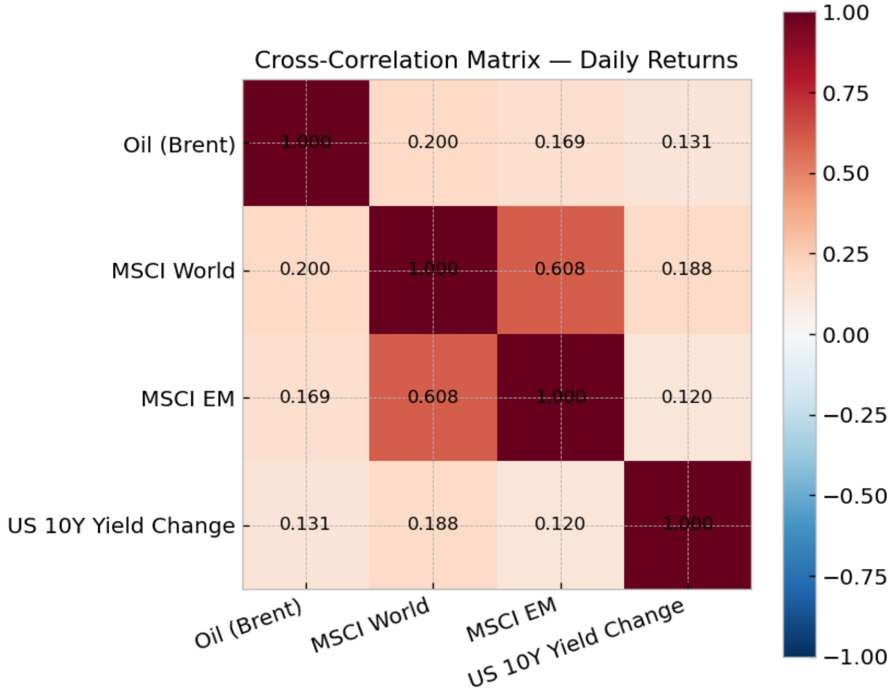

Table 1: OLS Summary

All baseline OLS models — key coefficients

	Model	Coef	beta	t-stat	sig	N	R2
0	Baseline World (daily)	roil	0.0765	6.1760	***	9442.0000	0.0666
1	Baseline EM (daily)	roil	0.0790	6.0850	***	9442.0000	0.0382
2	Asymmetric World (daily)	oil_pos	0.0555	4.1730	***	9442.0000	0.0677
3	Asymmetric World (daily)	oil_neg	0.0948	3.6660	***	9442.0000	0.0677
4	Asymmetric EM (daily)	oil_pos	0.0589	3.9580	***	9442.0000	0.0389
5	Asymmetric EM (daily)	oil_neg	0.0965	3.4470	***	9442.0000	0.0389
6	Interaction World (monthly)	roil_monthly	0.0912	2.9240	***	434.0000	0.0649
7	Interaction World (monthly)	oil x recession	0.0597	0.5900		434.0000	0.0649
8	Interaction EM (monthly)	roil_monthly	0.1386	3.4930	***	434.0000	0.0622
9	Interaction EM (monthly)	oil x recession	0.0841	0.6010		434.0000	0.0622

Table 2: ADF and KPSS Summary

ADF and KPSS Summary

	ADF Stat	ADF p-val	ADF Stat?	KPSS Stat	KPSS p-val	KPSS Stat?
Oil (Brent)	-22.8277	< 0.001	Yes	0.0453	0.1000	Yes
MSCI World	-22.8671	< 0.001	Yes	0.0893	0.1000	Yes
MSCI EM	-15.6772	< 0.001	Yes	0.0626	0.1000	Yes
US 10Y Yield Change	-19.0854	< 0.001	Yes	0.1710	0.1000	Yes
Oil (monthly)	-16.8671	< 0.001	Yes	0.0495	0.1000	Yes
MSCI World (monthly)	-20.0494	< 0.001	Yes	0.0858	0.1000	Yes
MSCI EM (monthly)	-17.9006	< 0.001	Yes	0.0419	0.1000	Yes
CFNAI	-15.2059	< 0.001	Yes	0.1869	0.1000	Yes
Industrial Production	-2.0448	0.2673	No	2.5963	0.0100	No
Manufacturing ISM	-5.6506	< 0.001	Yes	0.1583	0.1000	Yes

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